

Multimodal and Multisensor Analytics for Identifying Anxiety Phases in Highly Anxious Individuals

1. Introduction

Anxiety is usually measured by asking people how they feel. Questionnaires like the HADS or STAI are well-validated and quick to administer, but they only capture what someone is willing and able to report. What they miss is the physiological side—heart rate shifts, changes in pupil size, differences in how long someone stares at a question before answering. These signals happen whether or not the person is aware of them.

Studies that do track physiology during anxiety assessment tend to use one sensor at a time. Heart rate is the most common. But heart rate goes up for a lot of reasons—exercise, caffeine, stress, excitement. A single modality can’t isolate anxiety from general arousal. The question I wanted to answer was whether combining multiple sensor streams with standard questionnaire data would produce a richer, more reliable picture of what anxiety actually looks like in the body.

To test this, I ran a longitudinal study with 10 participants over three weekly sessions. Each person completed the same battery of five psychological questionnaires while wearing an eye tracker (Pupil Labs Core), a chest-strap heart rate monitor (Polar H10+), and an electrodermal activity sensor (TEA GSR). Facial data was also captured through OpenFace. The analysis moves from individual-level baselines, through group-level trends, to multimodal correlation and unsupervised clustering.

2. Methods

2.1 Study Design

Parameter	Detail
Design	Longitudinal, within-subjects
Participants	10 adults (university cohort)
Sessions	3 per participant, one week apart
Session flow	Resting baseline → sensor calibration → questionnaire battery → debrief
Duration	~45–60 min per session

Keeping everything fixed across sessions—same questionnaire order, same room, same time of day where possible—was deliberate. Any changes in physiology across sessions can’t be explained by procedural differences.

2.2 Questionnaires

Five validated instruments were administered in the same order every session, totalling 88 questions per session:

Instrument	Items	What It Measures
Hospital Anxiety and Depression Scale (HADS)	14	Anxiety (7 items) and depression (7 items) severity
State-Trait Anxiety Inventory — State (STAI-S)	20	How anxious the person feels right now
State-Trait Anxiety Inventory — Trait (STAI-T)	20	How anxious the person generally tends to be
Big Five Inventory (BFI-10)	10	Personality traits (O, C, E, A, N)
Fear Questionnaire (FQ)	24	Phobia subtypes: agoraphobia, blood-injury, social

The HADS was the primary target for physiological alignment because its items directly probe anxiety and depression symptoms. The STAI provided the state vs. trait distinction, the BFI-10 captured personality context, and the FQ profiled specific phobias.

2.3 Sensors

Modality	Device	What It Records
Eye tracking	Pupil Labs Core	Gaze direction (x, y, z), pupil diameter, blink detection
Cardiac	Polar H10+	Heart rate (BPM), inter-beat intervals (IBI)
Electrodermal	TEA GSR	Skin conductance level, galvanic skin response
Facial	OpenFace (software)	Action units, head pose, gaze direction

All sensors recorded simultaneously. The Polar H10+ streamed over Bluetooth, the Pupil Labs Core captured gaze at its native rate, and the TEA GSR logged from the wrist. Each stream was timestamped so that physiological readings could be matched to specific questionnaire items after the session.

2.4 Data Processing

2.4.1 Heart Rate and HRV

Raw heart rate files (*hr_01–03.csv*) were cleaned by removing zero values and dropouts. Average HR per session was computed and compared against a resting baseline recorded at the start of each visit.

HRV was derived from the inter-beat interval files (*ibi_01–03.csv*). Two time-domain metrics were calculated per session using standard formulas:

Metric	Calculation	Anxiety Threshold
SDNN	Standard deviation of R-R intervals (ddof=1)	< 50 ms
RMSSD	Root mean square of successive IBI differences	< 20 ms

These thresholds follow Shaffer & Ginsberg (2017). SDNN below 50 ms or RMSSD below 20 ms indicates reduced autonomic regulation, which is associated with anxiety.

2.4.2 Eye Tracking

Raw gaze data (*sed_01–03.csv*) was preprocessed by renaming columns (*gazeDir.x/y/z* → *gaze_x/y/z*). Gaze velocity was computed as the Euclidean distance between successive gaze direction samples:

$$velocity = \sqrt{[(gaze_x_2 - gaze_x_1)^2 + (gaze_y_2 - gaze_y_1)^2 + (gaze_z_2 - gaze_z_1)^2]} / \Delta t$$

Fixations were identified wherever velocity fell below a threshold. Two thresholds appear in the pipeline: 0.01 for the cleaned fixation export (CSV_Fix), and 0.1 for the broader fixation analysis. Valid fixations were filtered to 150–300 ms duration. Blink detection used a threshold of 1.0 on the *leftEyeOpen* and *rightEyeOpen* columns—above 1.0 was counted as eyes-closed.

2.4.3 Response Timing

The psychometric app logged question presentation time and answer time for every item (*Psychometric_Test_Results_01–03.csv*). Answer duration was computed as the time difference between these two timestamps. Standard deviation of answer duration across items served as a measure of pacing variability—erratic timing can signal processing difficulty or indecision.

2.4.4 Multimodal Alignment

For each questionnaire item, the closest HRV reading (SDNN, RMSSD) was found by matching timestamps. Concurrent eye metrics (pupil dilation, blink rate) were extracted for the same window. This produced a per-question, per-session, per-participant feature vector that pairs what someone reported with what their body was doing at that moment.

3. Analysis

The analysis was structured in three layers: individual-level (one participant in depth), paired comparisons (People directory), and group-level aggregation across all 10 participants (Style directory). This layering matters because patterns that hold for one person don't always hold at the group level, and vice versa.

3.1 Individual-Level (DM)

One participant's data was analyzed in depth to develop and test the full pipeline before scaling to the group.

Heart Rate

Resting baseline HR averaged 63.74 BPM. During the questionnaire sessions, HR rose substantially:

Session	Average HR	Change from Baseline	Classification
Baseline	63.74 BPM	—	Normal
Session 1	96.01 BPM	+32.26 BPM	Anxiety (> 90 BPM)
Session 2	92.43 BPM	+28.68 BPM	Anxiety (> 90 BPM)
Session 3	93.38 BPM	+29.63 BPM	Anxiety (> 90 BPM)

That's a 45–51% increase from rest to testing—not a subtle shift. A within-session paired t-test comparing early vs. late testing HR showed a 3.22% decline over the course of a single session ($p = 6.92 \times 10^{-10}$), suggesting participants calmed slightly as the session progressed, but never returned to resting levels.

HRV

HRV told a more nuanced story. Baseline and all three sessions showed RMSSD above the 20 ms anxiety threshold (no anxiety by that metric), but SDNN stayed below 50 ms across every session:

Session	RMSSD	SDNN	RMSSD Status	SDNN Status
Baseline	30.08 ms	28.63 ms	Normal	Anxiety
Session 1	42.83 ms	36.23 ms	Normal	Anxiety
Session 2	26.76 ms	29.91 ms	Normal	Anxiety
Session 3	36.53 ms	30.75 ms	Normal	Anxiety

This split—RMSSD normal, SDNN reduced—is informative. RMSSD reflects short-term (beat-to-beat) parasympathetic activity, while SDNN captures overall variability including both sympathetic and parasympathetic input. A participant can have adequate short-term vagal tone but still show reduced overall variability, which is exactly what happened here.

Fixation

Fixation analysis revealed a clear baseline-to-testing shift. Using a gaze velocity threshold of 0.1, baseline fixation duration averaged 174.73 ms—below the 200 ms lower bound of the normal range. By Session 2, fixation duration climbed to 543.56 ms, well within normal. Sessions 1 (198.30 ms) and 3 (207.74 ms) fell near the boundary. The short baseline fixations likely reflect the novelty of the setup—more scanning, shorter dwells.

HRV During Individual HADS Items

By aligning HRV timestamps to individual HADS questions, I could see which items triggered autonomic changes. Some anxiety-specific items (as opposed to the depression items) consistently produced drops in SDNN and RMSSD, suggesting that question content—not just the act of answering—drives part of the physiological response.

3.2 Paired Participant Analysis (People)

The People directory extended the analysis to paired comparisons. Key findings:

HR correlations across sessions. Pearson correlations between baseline, Session 1, Session 2, and Session 3 HR were low ($r = 0.23$ between Baseline and Session 1, $r = -0.12$ between Session 1 and Session 3). In other words, HR in one session didn't predict HR in the next.

Question-level HRV. RMSSD and SDNN were computed for each of the 14 HADS questions per session. Early questions (Q1–Q2) showed elevated HRV (RMSSD ~72 ms, SDNN ~67 ms), while later questions dropped sharply (Q5: RMSSD = 10.86 ms, SDNN = 7.53 ms). Questions that triggered HRV above the session baseline were flagged as “increase” events.

Eye metrics per question. Pupil dilation, left blink rate, and right blink rate were tracked for each HADS item. Sample values: Q1 pupil = 3.14 mm with 4.72 blinks/min, Q4 pupil = 2.92 mm with 0 blinks/min. The combination of pupil size and blink frequency varied substantially across questions.

3.3 Group-Level Analysis (Style)

The Style directory pulled together all 10 participants across all 3 sessions. This is where the clearest patterns emerged.

Group HRV (SDNN)

SDNN was calculated from IBI data for every participant and session:

Participant	Session 1	Session 2	Session 3
P01	80.14 ms	65.39 ms	1012.69 ms *
P02	46.20 ms	47.81 ms	47.40 ms
P03	54.98 ms	56.83 ms	39.02 ms
P04	41.38 ms	39.89 ms	55.02 ms
P05	34.73 ms	32.62 ms	37.63 ms
P06	41.64 ms	122.35 ms	117.05 ms
P07	42.21 ms	59.67 ms	50.04 ms
P08	188.91 ms	56.44 ms	58.87 ms
P09	38.50 ms	41.89 ms	47.27 ms
P10	43.43 ms	50.13 ms	55.75 ms

P01 Session 3 (1012.69 ms) is almost certainly a sensor artifact—likely a loose chest strap or Bluetooth dropout. P08 Session 1 (188.91 ms) is also unusually high. Both are flagged but kept in the dataset because with only 10 participants, removing data points requires strong justification.

Most participants fell between 30–70 ms. Using the 50 ms threshold, roughly half the participant-sessions show SDNN below that line—consistent with reduced autonomic regulation during testing. P06 showed a notable jump from Session 1 (41.64 ms) to Sessions 2–3 (117–122 ms), which could reflect genuine physiological change or a sensor calibration difference.

Pupil Dilation Variability

Standard deviation of pupil diameter was computed per person per session. Reference lines were set at 0.1 (normal), 0.2 (moderate), and 0.3 (high variability). However, actual participant values ranged from 0.38 to 1.25—every single participant exceeded the “high” reference in every session. This suggests the reference thresholds may be calibrated for different conditions or equipment, or that psychometric testing inherently produces high pupil variability compared to resting norms. The more useful finding is relative: each participant’s pupil variability stayed roughly consistent across sessions.

Response Timing

Average answer duration and its standard deviation were plotted for all 10 participants across sessions. Some got faster by Session 3 (a practice effect), but their relative ranking in the group barely changed—the slowest responder in Session 1 was still among the slowest in Session 3. This mirrors the HRV finding individual patterns persist even as absolute values shift.

Cross-Modal Correlation

A Pearson correlation matrix was computed across HRV SDNN, pupil dilation STD, and response duration STD for all 10 participants and all 3 sessions. The correlations were weak—cardiac and ocular metrics showed little shared variance. This is actually important: it means these three modalities are capturing different aspects of the physiological response. Heart rate variability indexes vagal tone, pupil dynamics index sympathetic arousal and cognitive load, and response timing indexes decision-making processes. If they were highly correlated, recording multiple sensors would be redundant.

3.4 Clustering

After combining all modalities into a single feature matrix, I applied PCA to reduce the space to two components, then ran K-Means with $k = 2$ through 10. Three evaluation metrics guided the choice:

Metric	What It Measures	Best k
Silhouette Score	How well each point fits its cluster vs. the next closest	3
Davies-Bouldin Index	Average cluster similarity (lower = better separation)	3
Calinski-Harabasz Index	Between-cluster vs. within-cluster variance (higher = tighter)	3

All three converged on $k = 3$. Features were standardized with StandardScaler before PCA. When I cross-referenced the three clusters against HADS scores, they corresponded to low, moderate, and high anxiety profiles—the physiology was grouping people the same way the questionnaire did, without being told the scores.

4. Results

4.1 Main Finding

The central result is a contrast: high variability between people, low variability within each person. Every participant had a distinct physiological signature—their own pattern of HRV, pupil dynamics, and response timing. But that signature stayed remarkably stable across the three sessions. P02’s SDNN barely moved (46–48 ms across all three sessions). P05 stayed in a tight band (33–38 ms). Even participants with higher absolute values (P06, P08) showed internal consistency once outlier sessions were accounted for.

This matters because it suggests these physiological patterns reflect stable traits, not momentary state. A baseline calibration done once could remain valid for weeks.

4.2 Anxiety Detection Thresholds

Metric	Normal Range	Anxiety Threshold	Notes
Heart rate	≤ 90 BPM	> 90 BPM	Individual analysis; group uses > 100 BPM
SDNN	> 50 ms	< 50 ms	Shaffer & Ginsberg (2017)
RMSSD	> 20 ms	< 20 ms	Shaffer & Ginsberg (2017)
Fixation duration	200–600 ms	< 200 or > 600 ms	Velocity-based detection
Pupil dilation	< 3.5 mm	> 3.5 mm	Absolute diameter
Blink rate	Baseline-referenced	> 1.0 (closed threshold)	Applied to leftEyeOpen/rightEyeOpen

4.3 Cluster Profiles

Cluster	Profile	Characteristics
1	Low anxiety	SDNN > 50 ms, stable fixation (200–400 ms), low pupil variability, HADS scores in normal range
2	Moderate	Borderline SDNN (30–50 ms), variable fixation, moderate pupil shifts, HADS borderline
3	High anxiety	SDNN < 30 ms, very short or very long fixations, high pupil variability, elevated blink rate, higher HADS scores

The alignment between unsupervised clusters and self-reported HADS scores is the strongest validation in this study. The clustering algorithm had no access to questionnaire scores—it grouped people based on physiology alone, and the groups matched the questionnaire results.

5. Discussion

The core argument for multimodal recording comes from the weak cross-modal correlations. Heart rate variability, pupil dynamics, and response timing are driven by partially independent autonomic and cognitive pathways. Using one sensor alone means missing the parts of the anxiety response that show up in the other channels. The three-cluster result from PCA + K-Means demonstrates that combining modalities captures structure that no single measure reveals on its own.

The within-person stability across sessions is probably the most practically useful finding. If someone’s physiological profile barely changes over three weeks, a one-time calibration session

could establish a reliable individual baseline. Future systems wouldn't need to recalibrate every visit—the person's signature persists.

The individual HRV data tells a more complex story than simple threshold classification. The index case showed RMSSD consistently above 20 ms (no short-term vagal deficit) but SDNN consistently below 50 ms (reduced overall variability). This dissociation—adequate parasympathetic tone but compressed total HRV—isn't captured by a single threshold check. It suggests that anxiety manifests differently in different HRV metrics, and that using both RMSSD and SDNN together gives a more complete picture than either alone.

The HR data adds another dimension. Resting HR of 63.74 BPM climbing to 92–96 BPM during testing is a substantial activation. But the within-session decline of 3.22% ($p = 6.92 \times 10^{-10}$) shows that participants gradually settled during the questionnaire—they started activated and slowly came down, though never back to resting levels. This temporal pattern would be invisible to a single-timepoint measurement.

5.1 Limitations

Ten participants is small. The patterns are consistent and the clustering is clean, but generalization to clinical populations, older adults, or different cultural contexts would require a larger and more diverse sample.

Two data points need to be flagged: P01's SDNN of 1012.69 ms in Session 3 and P08's 188.91 ms in Session 1. Both are likely sensor artifacts. I kept them because with $n = 10$, removing points changes group statistics more than it clarifies them. An automated artifact rejection pipeline would handle this in future work.

The pupil dilation reference thresholds (0.1/0.2/0.3 for normal/moderate/high) didn't match the observed data—all participants exceeded 0.3 in every session. These thresholds may be appropriate for resting-state or non-task conditions but not for active psychometric testing. Task-specific norms would need to be established.

OpenFace data was collected (facial action units, head pose) but not fully incorporated into the analysis. This represents a fourth modality that could refine the clustering.

5.2 Next Steps

A larger sample with clinically diagnosed anxiety disorders is the clear next step. If the three-cluster structure holds with more participants and a wider severity range, the thresholds become clinically actionable. Real-time classification during testing—adjusting which questions to ask based on physiological state—is another direction. The OpenFace data is ready to be integrated whenever the pipeline is extended. And the question-level HRV analysis (where specific HADS items trigger autonomic changes) could feed into item-level adaptive testing.

6. Conclusion

Simultaneous recording from multiple sensors during standardized psychometric testing produces stable, person-specific physiological profiles that align with self-reported anxiety. Three anxiety severity clusters emerged from unsupervised learning on combined cardiac, ocular, and behavioral features—without any access to questionnaire scores. Individual profiles stayed consistent across three weekly sessions, pointing to stable traits rather than session noise. The weak correlations between modalities confirm that each sensor captures a different piece of the anxiety response, making the case for multimodal recording over single-sensor approaches.

7. References

1. Zigmond, A. S., & Snaith, R. P. (1983). The Hospital Anxiety and Depression Scale. *Acta Psychiatrica Scandinavica*, 67(6), 361–370.
2. Spielberger, C. D. (1983). *Manual for the State-Trait Anxiety Inventory (STAI)*. Consulting Psychologists Press.
3. Rammstedt, B., & John, O. P. (2007). Measuring personality in one minute or less. *Journal of Research in Personality*, 41(1), 203–212.
4. Marks, I. M., & Mathews, A. M. (1979). Brief standard self-rating for phobic patients. *Behaviour Research and Therapy*, 17(3), 263–267.
5. Shaffer, F., & Ginsberg, J. P. (2017). An overview of heart rate variability metrics and norms. *Frontiers in Public Health*, 5, 258.
6. Holmqvist, K., et al. (2011). *Eye Tracking: A Comprehensive Guide to Methods and Measures*. Oxford University Press.
7. Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.